

Open Source for Home Use

3.1 Introduction

You have got the latest gear and your CPU is a water-cooled over-clocked monster satisfying your urge to live on the edge, but your software is holding you back. Either the latest and greatest lines of code are available as just “lines of code” or the available binary packages are designed for a generic environment and are not fully using the bleeding edge features of your machine. What to do? This chapter aims to provide the answer by teaching you how to compile software’s source code and install it on your machine.

As briefly mentioned in the box there are various distinct advantages of compiling the source code on your own, instead of using pre-built binary packages (exe/msi etc for Windows and deb/rpm for GNU/Linux) which are as follows:

Availability: It massively increases the range of software available for you. A lot of new and interesting software are available only as source archives, especially the ones in beta stages.

Optimisation: When you compile from source code on your own machine, the software is optimised for your specific hardware and will perform better and faster as compared to the generic built.

Bleeding edge: The daily builds of even mature software or the

Jargon buster

Before we move further it will be useful to go over a few definitions:

Source code: It is the actual human-readable code written in a language such as C, C++ or Java which is then compiled and linked by a software called “Compiler” and “Linker” (often rolled into one) to form an executable (your typical “filename.exe” file in Windows). This code can be modified to introduce, remove or modify desired features.

Binary package: This is your “setup.exe” or “filename.deb” or “filename.rpm” and similar software installer. This is machine-readable instructions and do not make sense to human eyes. Binary packages cannot be modified.

Open source software: People who write these software willingly share the source code of the same with everyone. Their prime motive to share their intellectual property with others is their belief that this open collaboration produces better results and prevents people from solving the same problem twice.